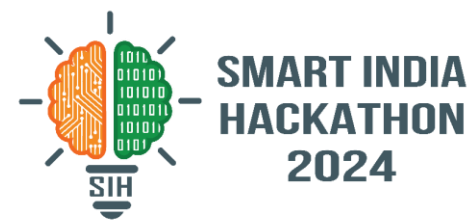


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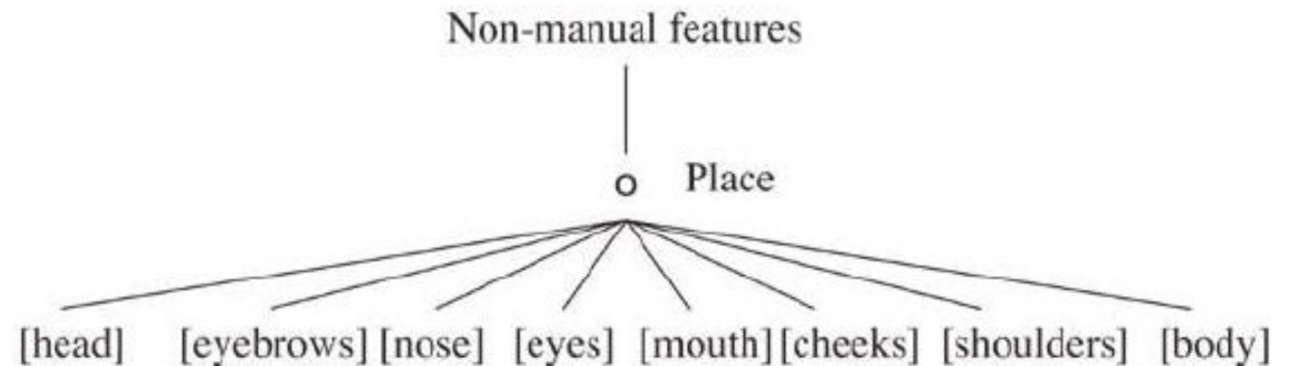
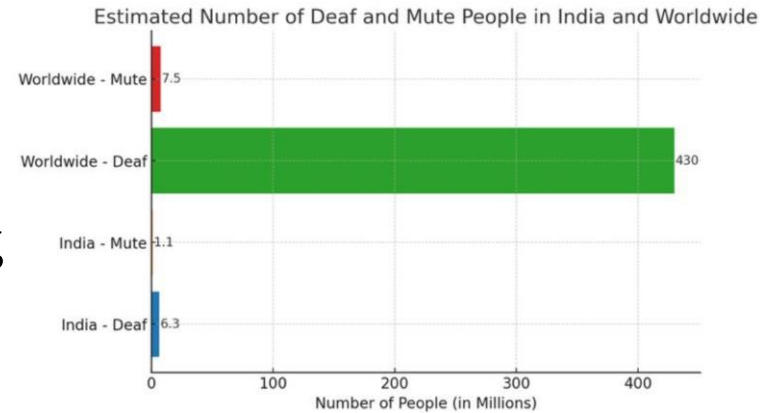


- **Problem Statement ID- SIH1718**
- **Problem Statement Title- Capturing Non-manual features of Indian Sign Language & converting it into text**
- **Theme- Miscellaneous**
- **PS Category- Software**
- **Team ID- INT78**
- **Team Name- The Hacktivists**



✓ Proposed Solution

- Sign Language Translator using machine learning for hand gesture recognition along with non-manual gesture detection.
- It translates sign language into text or speech, aiding communication for the hearing impaired, bridging communication gaps between sign language users and non-users.
- Ensuring with the proposed solution that these individuals get equal chances for employment, education and civil rights.



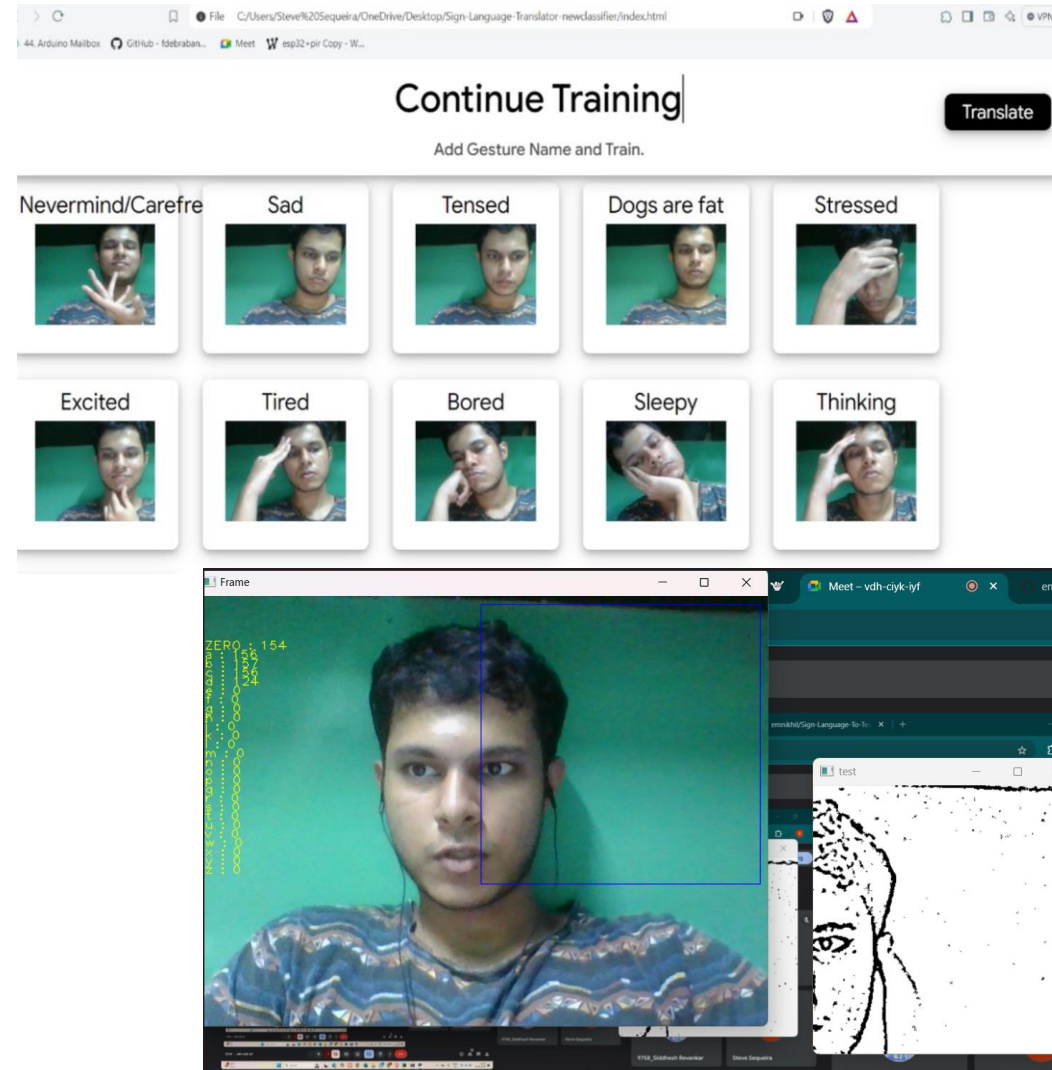
Technology Stack:

Software requirements:

- Frontend - React, HTML, CSS
- Backend - Python, NodeJS
- Database - PostgreSQL/MongoDB
- ML Libraries - tensorflow/pytorch
- Computer vision - OpenCV
- Real time processing - Websocket

Hardware requirements:

- Camera
- Laptop/Desktop
- Networking and Connectivity: Wi-Fi



Data Feasibility:

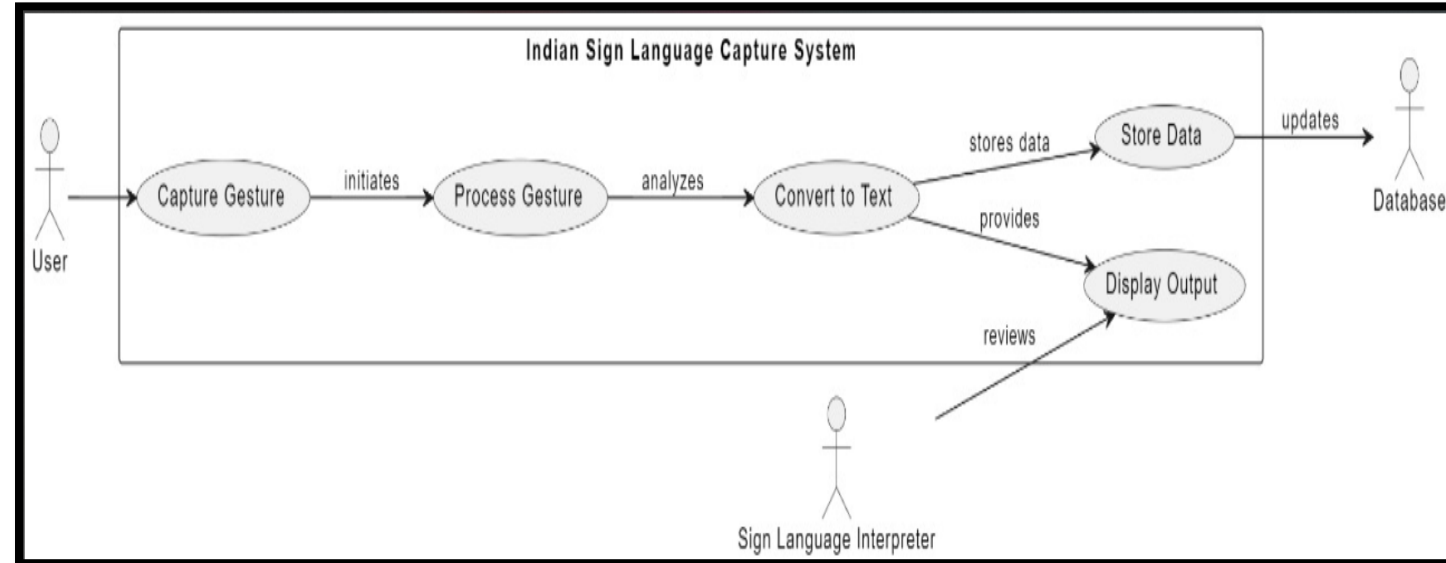
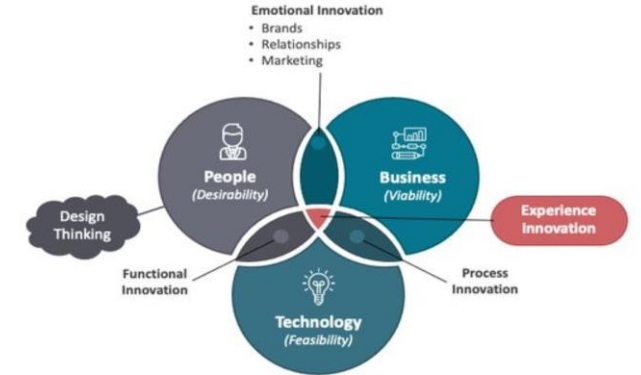
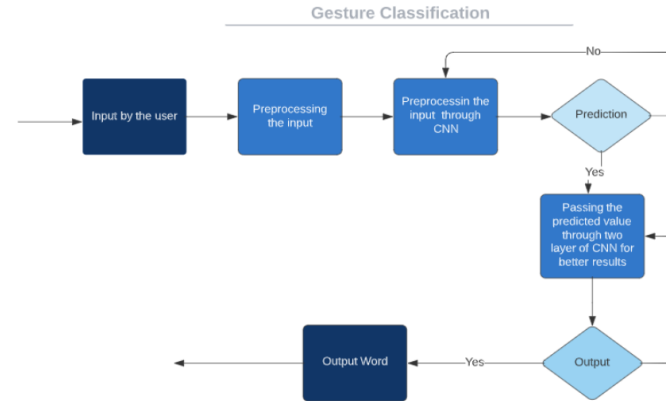
Accurate gesture data collection is challenging, requiring extensive datasets to cover diverse sign languages and gestures.

Risks and Feasibility:

Technical risks include gesture recognition accuracy and real-time processing limitations.

Strategies to Overcome These Challenges:

Optimization (Enhancing model accuracy and performance)



Novelty:

1. Incorporating regional and cultural variations of ISL, ensuring inclusivity.
2. AI based gesture prediction: Developing an AI that can predict the upcoming signs and make translation more fluid and dynamic.
3. Offline mode availability: Making the application to function offline, hence ensuring availability in remote areas and also with limited or no internet connectivity.

Benefits:

- Better Communication
- Clear Understanding
- Increased Accessibility
- Learning Aid



SDGs Mapped



- Indian Sign Language recognition system using SURF with SVM and CNN [2022]:
<https://www.sciencedirect.com/science/article/pii/S2590005622000121>
- Deep learning-based sign language recognition system using both manual and non-manual components fusion [2023]: <https://www.aimspress.com/aimspress-data/math/2024/1/PDF/math-09-01-105.pdf>
- Deep Learning for Sign Language Recognition: Current Techniques, Benchmarks, and Open Issues [2021]: <https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=9530569>
- OpenHands: Making Sign Language Recognition Accessible with Pose-based Pretrained Models across Languages [2021]: <https://arxiv.org/pdf/2110.05877>
- Multi-Semantic Discriminative Feature Learning for Sign Gesture Recognition Using Hybrid Deep Neural Architecture [2022]: <https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=10004966>
- Sign language recognition using the fusion of image and hand landmarks through multi-headed convolutional neural network [2023]: <https://www.nature.com/articles/s41598-023-43852-x.pdf>
- A Review of Hand Gesture Recognition Systems Based on Noninvasive Wearable Sensors [2023]:
<https://onlinelibrary.wiley.com/doi/pdfdirect/10.1002/aisy.202300207>